

Examiner Report

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Anmelde-Nr.
Application No: 16 777 760.6
Demande n°:

The examination is being carried out on the **following application documents**

Description, Pages

1-22 filed with entry into the regional phase before the EPO

Claims, Numbers

1-14 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/12-12/12 filed with entry into the regional phase before the EPO

The following document has been cited in the international search report; the numbering will be adhered to in the rest of the procedure:

D6 GB 2 516 335 A (GW PHARMA LTD [GB]) 21 January 2015 (2015-01-21)

The following documents are cited by the examiner. A copy of the documents is annexed to the communication and the numbering will be adhered to in the rest of the procedure:

D12 US 2014/302121 A1 (BEVIER JONATHAN [US]) 9 October 2014 (2014-10-09)

D13 EP 1 559 423 A1 (NL ORGANISATIE VOORTOEGEPAST N [NL]) 3 August 2005 (2005-08-03)

D14 WO 2008/129258 A1 (GW PHARMA LTD [GB]; DI MARZO VINCENZO [IT] ET AL.) 30 October 2008 (2008-10-30)

The third-party observation submitted on 31-05-2019 makes reference to Exhibits 1-7. A copy of Exhibits 2,4,6 and 7 is annexed to this communication and the numbering will be adhered to in the rest of the procedure:

Exhibit 2 "Vijaya Kalpah", TKDL, 1 January 1952 (1952-01-01), XP003029556

Exhibit 4 DATABASE GNPD [Online] MINTEL; 1 January 1952 (1952-01-01), anonymous: "Vijayeovararasaa", XP055625445, retrieved from www.gnpd.com Database accession no. RS/2047

Exhibit 6 DATABASE GNPD [Online] MINTEL; 1 January 1921 (1921-01-01), anonymous: "Raughan Bara-e- Chhajan", XP055625453, retrieved from www.gnpd.com Database accession no. MH1/3868

Exhibit 7 DATABASE GNPD [Online] MINTEL; 24 September 2019 (2019-09-24), anonymous: "Kodivaeli Chooranam", XP055625459, retrieved from www.gnpd.com Database accession no. VP01/70

1. Clarity, support and disclosure (Articles 83 and 84 EPC)

The application does not meet the requirements of Articles 83 and 84 EPC, because claims 1-4 are not clear nor sufficiently supported and the invention is not sufficiently disclosed by the description.

1.1. With entry into the regional phase, the claims have been restricted to cannabidiolic acid (CBDA) for use in the treatment of an inflammatory skin disease selected from a microbial-induced dermatitis, solar dermatitis and atopic dermatitis (see claim 1).

However, dependent claims 2-4 cover numerous pathological conditions which are not falling within the scope of claim 1, as they encompass diseases other than dermatitis.

Moreover, although some of the diseases mentioned in claim 2 may have an inflammatory component, in general a number of said diseases cannot even be considered to be inflammatory skin diseases.

Therefore, there is an inconsistency between the extent of independent claim 1 and the extent of claims 2-4 which casts a doubt on the subject-matter actually covered by the claims (Article 84 EPC).

1.2. Furthermore, the most of the therapeutic applications according to claims 2 and 4 are not sufficiently supported (Article 83 EPC). These claims cover many different skin diseases, however, the application only provides experimental data for a very limited number of pathological conditions, i.e. those possibly linked to microbial infection-induced dermatitis (Table 1), solar induced dermatitis (Table 2), and atopic dermatitis (Table 3).

Based on the results provided in Table 1 it is plausible that CBDA would be effective for the treatment of infection-induced dermatitis. However, looking at the low scores shown for CBDA in Tables 2 and 3, it appears that the claimed compound would not be particularly effective for the treatment of solar induced dermatitis or atopic dermatitis.

It appears that the effect demonstrated microbial infection-induced dermatitis could not be generalized to all of the (inflammatory) skin diseases mentioned in claim 1-4, such as for example skin cancer, acne, verrucas, herpes, scabies, fungal infections, rosacea, psoriasis or alopecia just to mention some.

2. Novelty (Article 52(1) and 54 EPC)

The present application does not meet the requirements of Article 52(1) EPC, because the subject-matter of claims 1-9,11-14 is not new in the sense of Article 54(1) and (2) EPC).

2.1. D6 (paragraphs [0001], [0013] - [0026], [0030] - [0032], [0036], [0067] - [0077]; claims 1-9) discloses the use of phytocannabinoids selected from CBD, CBDA, CBG, CBGV and THCV, for the treatment of basal cell carcinoma. The cannabinoids can be used as an extract, in an isolated or pure form, or they can be synthetic. Example 3 shows the cytotoxic effect of the combination CBD+CBG in a model of skin carcinoma. With regard to CBDA for use in the treatment of CBDA, D6 and present application provide the same level of technical teaching

D6 is considered to be prejudicial to the novelty of claims 1,3,6-10,12,14-19.

2.2. D12 (paragraphs [0079], [0092], [0093], [0106], [0137] - [0164]; claim 12) discloses cannabinoid receptor binding agents, including CBDA, for the treatment of a dermatological inflammation, skin cancer (including basal cell carcinoma and squamous cell carcinoma), psoriasis, pruritus, alopecia, and skin inflammation caused by Tinea sp.

D12 is considered to be prejudicial to the novelty of claims 1,2,4,7,8,9,11-14.

D13 (paragraphs [0001], [0010], [0013], [0025], [0036]; claims 1,9,10,18) discloses acidic cannabinoids, including cannabidiolic acid (CBD-A, also referred to as CBD when used in a mixture with cannabidiol) for the treatment of inflammatory skin diseases such as dermatitis and psoriasis.

D13 is considered to be prejudicial to the novelty of claims 1-5,11-14.

D14 (page 1, lines 5-25; page 9, lines 1-11, 24-32; page 11, lines 13-15, 26-31; page 12, lines 24-26; page 13, lines 1-5,21-27; page 14, lines 23-26,30; page 15, lines 1-4,25; page 19, line 1 - page 20, line 17; claims 1,4,8,15) discloses the use of cannabinoid-containing plant extracts, also containing CBDA as the predominant cannabinoid, inter alia for the treatment of inflammation, cancer of the skin and shingles, via blockade of TRP channels.

D14 is considered to be prejudicial to the novelty of claims 2,5-8. In case the above pathologies were considered to fall under the meaning of inflammatory skin disease according to claims 1 and 14 (see item 1.1. above), then this document is also prejudicial to claims 1,11-14.

2.3. The third-party observation submitted on 31-05-2019 makes reference to Exhibits 1-7, all sourced from the Traditional Knowledge Digital Library (TKDL).

Exhibits 1, 3 and 5 (not annexed to this communication) are not relevant since they relate to conditions/treatments which are not encompassed by present application, i.e. swelling, sweating, complexion promoting.

Exhibits 2, 4, 6 and 7 are considered to be relevant. They all relate to therapeutic compositions of the traditional knowledge comprising parts of Cannabis sativa useful for the treatment of eczema and psoriasis (Exhibits 2 and 6), vitiligo (exhibit 4), and alopecia (Exhibit 7).

It is well known in the art that CBDA is contained in Cannabis plant together with other cannabinoids (see e.g. D6, paragraphs [0026] and [0030]; D14, page 9, lines 24-31). However, Exhibits 2, 4, 6 and 7 do not specify which of the cannabinoids contained in the Cannabis sativa parts used in the composition disclosed therein are responsible for the therapeutic/cosmetic effect disclosed. Therefore said documents do not appear to be prejudicial to the novelty of independent claim 1, which is interpreted as requiring that CBDA is the active compound actually responsible for the claimed effect.

However, Exhibits 2 and 6 are considered to be prejudicial to the novelty of claims 11-14 since they indeed disclose compositions comprising CBDA for use in the treatment of eczema. The same applies for Exhibits 4 and 7 in case vitiligo and alopecia were to be considered as falling under the meaning of inflammatory skin diseases according to independent claim 11 (see item 1.1. above).

3. Inventive step

The present application does not meet the criteria of Article 33(3) PCT, because the subject-matter of claims 1-14 does not involve an inventive step.

3.1. The problem to be solved by present application is to provide a treatment for inflammatory skin diseases according to claims 1-4 and 11.

The proposed solution is the use of cannabidiolic acid (CBDA).

The use of CBDA in the treatment of skin diseases according to the present application is already known from D6, D12-D14 and Exhibits 2, 4, 6 and 7, thus, even in the event that novelty could be restored, an inventive step still need to be demonstrated over said documents.

3.2. Moreover, claim 1 covers any type of inflammatory skin diseases, encompassing a large number of possible pathological conditions. The application however only provides a demonstration of an effect on a very limited number of pathological conditions, i.e. those possibly linked to microbial infection-induced dermatitis (Table 1), solar induced dermatitis (Table 2), and atopic dermatitis (Table 3).

Furthermore, based on the results provided in Tables 1-3, and in particular the low scores shown for CBDA in Tables 2 and 3, it would appear that CBDA would be not particularly effective for the treatment of solar induced dermatitis or atopic dermatitis, i.e. would not solve the problem posed.

Also, it appears that the effect demonstrated could not be generalized to all of the inflammatory skin diseases according to the claims. As a rule, protection conferred by a patent should be commensurate with the subject-matter which has been adequately demonstrated, including obvious variants thereof. However, it appears that the ambit of present claims is not commensurate with the contribution to the art.

3.3. Claims directed to CBDA for use in the treatment of a microbial infection-induced dermatitis would be considered both novel and inventive over the prior art. The closest prior art document would be D12, which disclose the use of cannabinoids, including CBDA, for the treatment of *inter alia* skin inflammation caused by *Tinea* sp. However, Table 1 of present application shows that CBDA exerted a better effect compared to the other cannabinoids in the TLR3-induced inflammation model, which mimics microbial infection-induced dermatitis. This effect is surprising and could not be expected based on the disclosure of D12, neither alone or in combination with any of the documents on file.